

ECE 315

Steven Knudsen, PhD, PEng

Lecture 1 :

- Course Introduction



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This Lecture

Introduction

- Core course information
- Course objectives
- Overview of content
- Assessment, grading, other policies

Course overview

- What this course about
- Where does it fit in the ECE and MCTR programs

Managing your work

- Git and Github
- Canvas

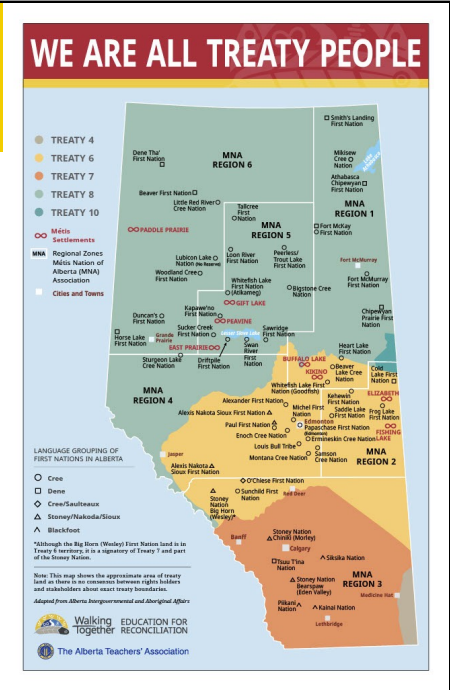
First Concepts

- What is a real-time embedded system?
- Why is computer interfacing important?

My Treaty Acknowledgement

My home and workplace at the University of Alberta are located in ᑭᓴᓂᗵᓐᓄᓐᓇᓱᓐ (Amiskwaciwâskahikan) on Treaty 6 territory, the territory of the Papaschase, Nêhiyaw, Dene, Anishinaabe, Nakota Isga, and Niitsitapi peoples, homeland of the Métis Nation in Region 4, and home to one of the largest Inuit community south of the 60th parallel.

My family are on Treaty 7 territory, the traditional territories of the Blackfoot Confederacy (Siksika, Kainai, Piikani), the Tsuut'ina, the Iyāxe Nakoda Nations, and the homeland of the Métis Nation in Region 3.



Instructor Information

Steven Knudsen, PhD, PEng (he/him/his)

- U of A Computer Engineering 1984
- U of A Applied Sciences in Medicine 1987
- U of C Electrical Engineering 1993
- 35+ years in industry
- E.I.T to VP to CTO & all between
- Mostly communications
- Commercial, military, space
- Office : DICE 5-261 (access locked at 5 pm)
- Office hours by appointment and/or via Zoom/Meet
- (may try) Canvas Discussions for messaging
- Will attend all labs except some Fridays because of ECE 492



Instructional Team

Antonio Andara Lara	Laboratory Instructor	andarala@ualberta.ca
Shyama Gandhi	Teaching Assistant (Lab)	smgandhi@ualberta.ca
Avik Mandal	Teaching Assistant (Lab)	amandal1@ualberta.ca
Muaz Shaikh	Teaching Assistant (Lab)	mohamad7@ualberta.ca
Suhani Mohanty	Teaching Assistant (Lec)	mohanty2@ualberta.ca
Junlong Shen	Teaching Assistant (Lec)	junlong6@ualberta.ca

Respectful Conduct

- Let me know if you want to be addressed by a name that is different than what is on Canvas.
- Please feel free to share your preferred pronouns.
- Please respect others' pronouns and use gender-neutral pronouns (they/them/theirs) when referring to people you don't know.
- As part of student accommodations, audio recordings may be made of lectures in accordance with UofA policies protecting your right to privacy

Course Information

- One lecture section, B1
 - Monday, Wednesday, and Friday, 1200-1250, CCIS 1-160
- Four lab sections
 - H21 Tuesdays 1400-1650 ETLC E5-001
 - H31 Wednesdays 1400-1650 ETLC E5-001
 - H41 Thursdays 1400-1650 ETLC E5-001
 - H51 Fridays 1400-1650 ETLC E5-001
- Quizzes
- No Midterm, but a Final
- Possibly bonus marks (more on that later)



Labs

- Teams of two people – ***must have a lab partner***
 - May choose your own in your section, or
 - One will be assigned
- First lab is mandatory
 - Easy marks
- Lab 5 will have a bonus section
 - Bonus marks added to lab mark; if lab mark maximized, marks added to assignments mark

Canvas and Syllabus

- We will make good use of Canvas
 - Full Lecture notes posted after lecture
 - Assignments and answers
 - Quizzes (after the fact) and answers
 - Pre-lab and In/Post-lab submissions
 - Discussion
 - Reference materials
 - HOWTOs, tutorials, examples
 - Links to online resources
 - List of recommended resources
- First year for course in this format
- Canvas resources will be populated as we progress
 - There may be mistakes, so please bear with us
 - Happy to receive feedback at any time, please email Steven

Use of AI – Instructor-Specified AI-Use

- In this course, we commit to AI use guided by ethical and transparent principles. While students are allowed to use advanced automated tools (such as ChatGPT, Gemini, Claude, or Dall-E 2) for certain written assignments, it is crucial to adhere to the following guidelines:
- Unless explicitly mentioned or written, seek prior approval from the instructor for AI use in specific assignments.

Use of AI – cont'd

- When allowed, clearly attribute and cite any AI-generated content in your work, including prompts and AI outputs as part of your academic record. Include an additional reflection component in your assessments, discussing how AI tools contributed to your learning process.
- What I expect is that in places provided you will say if you have used AI and what insights you have gained

Use of AI – cont'd

IMPORTANT

Please note that AI use is strictly prohibited in assessments and assignments not approved by the instructor.

Failure to abide by this guideline may be considered an act of cheating and a violation as outlined in the relevant sections of University of Alberta (April 2024) Code of Student Behaviour.



<https://www.ualberta.ca/en/governance/resources/policies-standards-and-codes-of-conduct/code-of-student-behaviour.html>



Our use of AI

- Assignments, Quizzes, and Labs : **AI is allowed**. Indicate, what, how, and where you have used it.
 - E.g., "I used ChatGPT to design a truth table for the Boolean expression in this assignment"
- **Plus**, you need to describe what insights it gave you. How did it help?
- Final : **AI is NOT allowed**. Phones and tablets will be placed face down on your desk. Not yet sure about using examination software...
- All other opportunities : Please discuss with the instructor

What to expect

- Combination of lectures, readings, and hands-on learning
- Use of simulations and physical devices
- Assignments, labs, quizzes, and a final
- A chance to earn bonus marks to bring up assignment and lab marks

This course provides knowledge & skills that will help your career

What to expect

- Lectures as PDFs posted after class
- Not all is captured in the slides and speaker notes.
 - For example, explanations of upcoming assignments, labs, quizzes
 - Any questions posed and answered in class
 - Other concepts, guidance, etc. I might provide
 - Please attend lectures!
- First year for using Canvas, first year for this version of the course
 - Canvas will be built out incrementally

Computer Engineering and Mechatronics

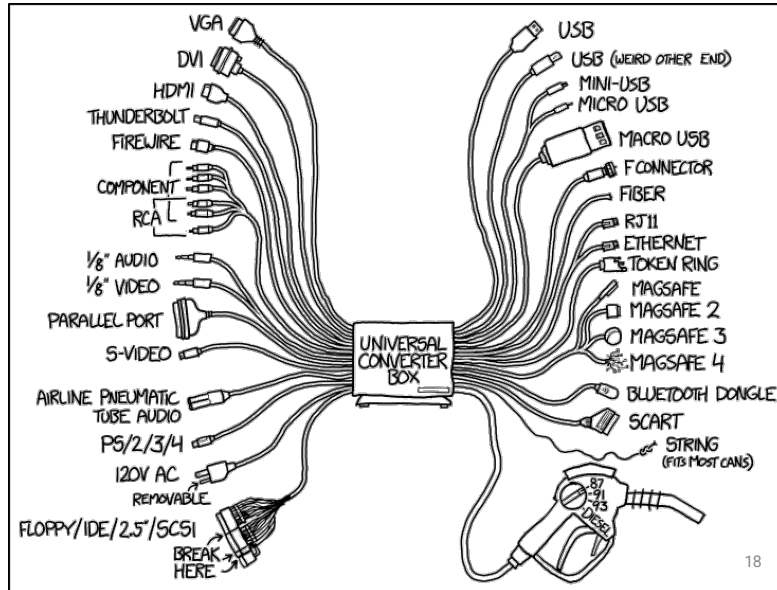
- This course is core to both programs
- It has been updated and has more MCTR material
 - Examples and labs are slightly skewed to Mechatronics systems and problem
 - Computer Engineers might expect to see such application or analogous
- Both programs are aimed at system-level understanding

First Ever... This Will not be Perfect

- This is the first time I teach ECE 315
 - Course has been changed based on previous students' feedback
- I have 40+ years experience, 35+ years in industry building real systems included deeply embedded
- We will learn together
- This is a career-foundational, we must make an effort to learn

ECE 315

Computer Interfacing?



<https://www.adafruit.com/2016/>

This lecture



Course learning objectives



Why are we here?



High-level course content



Schedule



Housekeeping

ECE 315 Learning Objectives

1. Explain the architecture and operation of embedded systems, including microcontrollers, peripherals, and I/O subsystems.
2. Differentiate between analog and digital interfacing methods used in embedded systems.
3. Design and implement digital input/output interfaces for sensors, actuators, and external devices.
4. Apply analog interfacing techniques using ADCs (Analog-to-Digital Converters) and DACs (Digital-to-Analog Converters).
5. Develop firmware to communicate with peripherals via standard protocols such as I²C, SPI, and UART.

ECE 315 Learning Objectives – cont'd

6. Use interrupt-driven programming and timers to manage real-time tasks in embedded applications.
7. Implement serial and parallel data communication between embedded devices and external systems.
8. Integrate sensors and actuators into a functional embedded system, handling signal conditioning and calibration.
9. Employ embedded software tools (e.g., IDEs, debuggers, compilers) to program and test hardware interfaces.
10. Demonstrate the ability to interface embedded systems with external devices such as displays, motors, and wireless modules.

ECE 315 Learning Objectives – cont'd

11. Gain experience with the use of the C programming language, including the use of pointers, to implement a real-time embedded system.
12. Become familiar with DC and stepper motor technology and be familiar with different ways of generating the digital signals that are required to control stepper motors.
13. Analyze and troubleshoot hardware–software integration issues using oscilloscopes, logic analyzers, and diagnostic tools.
14. Design, build, and document a embedded interfacing project that demonstrates real-world data acquisition or control.

ECE 315 Learning Objectives - Summary

- You should be comfortable thinking in terms low level operations with bits – getting them and outputting them
- You should have a good understanding of how a microprocessor/controller can be used to acquire real-world information, process it, and use the results to control other devices
- Should always ask what are the real timing requirements
 - Is some other machine relying on this? Probably hard real-time
 - Is a human relying on this? Probably soft real-time
 - Is this a safety-critical situation? Most definitely hard real-time!!!
- This will help you understand mechatronic and robotic systems

Why are we here?

- Well-established course in ECE
- Core course in MCTR and Computer Engg
- Similar to what I took long ago
 - Fundamental to my career
 - Used this knowledge at every stage
- CompE; ECE 212 is pre-req, ECE 315 is pre-req ECE 492
- MCTR; MCTR 210 is unofficial pre-req, ECE 315 should be pre-req for MCTR 460/461
- See visualizer



https://sites.ualberta.ca/~dnobes/Eng_Programs/MecTronE_Visualizer/index.html



High-level course content

- Reviews
 - Computer architecture
 - Operating systems
 - Tasks, threads, processes
 - Responding to events
 - Key C programming concepts
- Signals and the real world
- Responding to the real world
 - Hard/Soft Real-time
- Real-time Operating System
- FreeRTOS; intro and use
 - Labs
 - Assignments
- Embedded development
 - Example system we will use
- Sensors, motors, actuators
- Embedded communication
- Feedback control
- Specific processor architectures



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As in the syllabus

Schedule – Major Dates

- Quizzes

- #1 – Jan 23
- #2 – Feb 9
- #3 – Mar 6
- #4 – Mar 30

- Assignments

- #1a – Jan 14
- #1b – Jan 14
- #2 – Feb 4
- #3 – Mar 2
- #4 – Mar 23

- Final Exam

- synchronous
- TBC

- Labs – due following week

- #1 – Week 3 (Jan 19)
- #2 – Week 5 (Feb 2)
- #3 – Week 6 (Feb 23)
- #4 – Week 10 (Mar 9)
- #5 – Week 13 (Mar 23)

Git and Github Organization

Two tools used in ECE MCTR courses; introduced early

They help us

- Manage and track engineering work, including course work
- Submit assignments, labs, projects so they can be marked

Computer System Interfacing

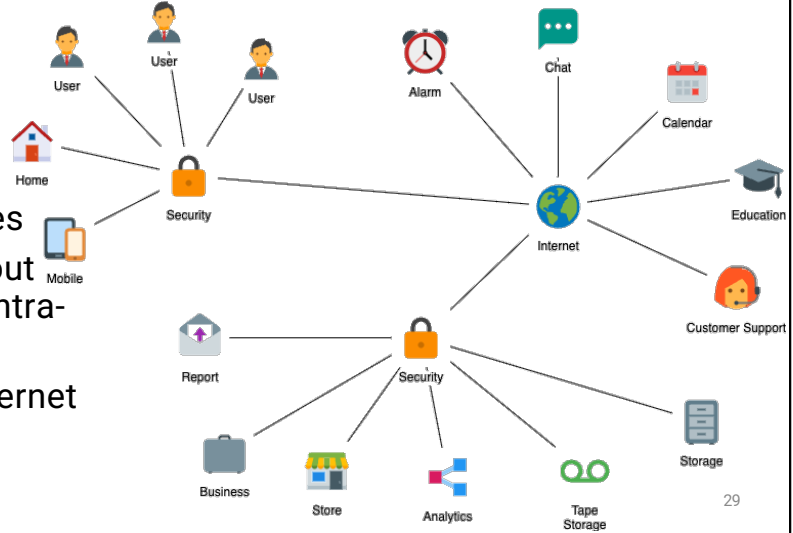


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What ECE 315 is not

- We are not interested in connecting personal, business, servers, clients, and other “big iron” machines
- This is not a course about networking, protocols, intra- or inter-nets
- We won't look at the Internet

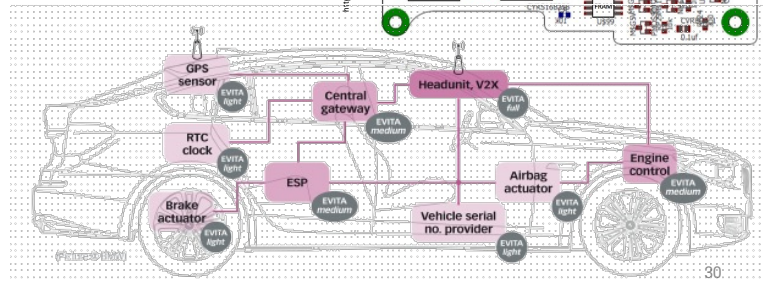
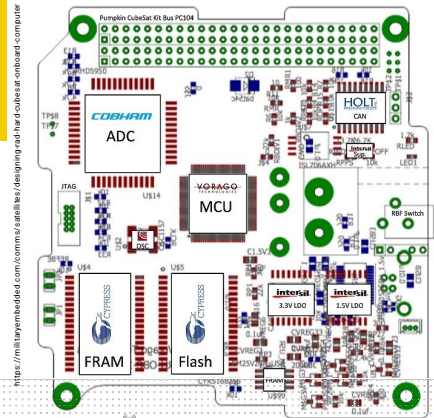


What ECE 315 is

- Connecting small, dedicated, embedded computers to the world
- "Headless" systems
- Hard- and Soft-real time requirements
- Low-cost
- Resource-constrained
- Low-power
- Small
- Billions of them!



Bouyahia, Tarek & Idrees, Sabir & Cuppens-Boulahia, Nora & Cuppens, Frédéric & Autrel, Fabien. (2015). Metric for Security Activities Assisted by Argumentative Logic. 8872. 10.1007/978-3-319-17016-9_12.



Next Lecture -

- Embedded Systems overview
 - Importance of embedded systems
 - Examples
- Challenges
 - SWaP
 - Constrained “everything”

